Identification Capacity and Rate-Query Tradeoffs in Classification Systems

Tristan Simas

Abstract

We study zero-error class identification under constrained observations with three resources: tag rate L (bits per entity), identification cost W (attribute queries), and distortion D (misidentification probability). If the attribute-profile map π is not injective on classes, then attribute-only observation cannot identify class identity with zero error. Let $A_\pi := \max_u |\{c : \pi(c) = u\}|$ be collision multiplicity. Any $D = 0$ scheme must satisfy $L \geq \log_2 A_\pi$, and this bound is tight. In maximal-barrier domains ($A_\pi = k$), the nominal point $(L, W, D) = (\lceil \log_2 k \rceil, O(1), 0)$ is the unique Pareto-optimal zero-error solution under the stated observation model: no other feasible combination achieves $D = 0$. Without tags ($L = 0$), zero-error identification requires $W = \Omega(d)$ queries, where d is the distinguishing dimension. Minimal sufficient query sets form the bases of a matroid, making d well-defined as an invariant of the observation structure rather than of a particular query strategy. We also prove exact finite-block confusability laws: at blocklength t, zero-error feasibility is equivalent to a tag alphabet threshold A_π^t, yielding exact linear log-bit scaling. Conceptually, the framework extends classical rate-distortion accounting by treating query cost as an additional interactive resource (structurally analogous to perception-constraint extensions). We give instantiations across databases, biology, software runtimes, and model registries, and machine-check the main claims in Lean 4 (8092 lines, 352 theorem/lemma statements, 0 `sorry`).

Keywords: rate-distortion theory, zero-error identification, confusability graphs, query complexity, matroid structure, classification systems

T. Simas is with McGill University, Montreal, QC, Canada (e-mail: tristan.simas@mail.mcgill.ca).

Supplementary artifact (Lean 4 formalization): https://doi.org/10.5281/zenodo.18123531

I. Introduction

A. The Identification Problem

Consider an encoder-decoder pair communicating about entities from a large universe $\mathcal{V}$. The decoder must *identify* each entity, determining which of k classes it belongs to, using only:

- A *tag* of L bits stored with the entity, and/or
- *Queries* to a binary oracle: "does entity v satisfy attribute I?"

This is not reconstruction (the decoder need not recover v), but *identification* in the sense of Ahlswede and Dueck [1]: the decoder must answer "which class?" with zero or bounded error. Our work extends this framework to consider the tradeoff between tag storage, query complexity, and identification accuracy.

We prove three core theorem families, and then extend them with finite-block, open-world, and mechanization results:

1) **Information barrier (identifiability limit).** When the attribute profile $\pi : \mathcal{V} \to \{0,1\}^n$ is not injective on classes, zero-error identification via queries alone is impossible: any decoder produces identical output on colliding classes, so cannot be correct for both.
2) **Optimal tagging (achievability).** A tag of $L = \lceil \log_2 k \rceil$ bits achieves zero-error identification with $W = O(1)$ query cost. For maximal-barrier domains ($A_\pi = k$), this is the unique Pareto-optimal point in the (L, W, D) tradeoff space at $D = 0$; in general domains, the converse depends on $A_\pi := \max_u |\{c : \pi(c) = u\}|$.
3) **Matroid structure (query complexity).** Minimal sufficient query sets form the bases of a matroid. The *distinguishing dimension* (the common cardinality of all minimal sets) is well-defined and lower-bounds the query cost W for any tag-free scheme.

These results are domain-independent within the stated observation model: the same lower bounds appear across databases, biological taxonomy, knowledge graphs, and software runtimes. Once classes collide under observable attributes, constant-cost zero-error identification requires auxiliary naming information. In maximal-barrier domains this yields a unique $D = 0$ Pareto solution, so the issue is resource feasibility, not domain-specific style conventions.

B. The Observation Model

We formalize the observational constraint as a family of binary predicates. The terminology is deliberately abstract; concrete instantiations follow in Section VII.

Definition I.1 (Entity space and attribute family)**.** Let $\mathcal{V}$ be a set of entities (program objects, database records, biological specimens, library items). Let $\mathcal{I}$ be a finite set of binary *attributes*. Each $I \in \mathcal{I}$ induces a bipartition of $\mathcal{V}$ via q_I, and the full family induces the observational equivalence partition.

Remark I.2 (Terminology). We use "attribute" for the abstract concept. Depending on domain, attributes may be interfaces/signatures, database columns, phenotypic characters, or catalog facets. The mathematics is identical.

Definition I.3 (Attribute observation family)**.** For each $I \in \mathcal{I}$, define the attribute-membership observation $q_I : \mathcal{V} \to \{0, 1\}$:

$$q_I(v) = \begin{cases} 1 & \text{if } v \text{ satisfies attribute } I \\ 0 & \text{otherwise} \end{cases}$$

Let $\Phi_{\mathcal{I}} = \{q_I : I \in \mathcal{I}\}$ denote the attribute observation family.

Remark I.4 (Notation for size parameters). We write $n := |\mathcal{I}|$ for the ambient number of available attributes. We write d for the distinguishing dimension (the common size of all minimal distinguishing query sets; Definition IV.9), so $d \leq n$ and there exist worst-case families with $d = n$. We write m for the number of *query sites* (call sites) that perform attribute checks in a program or protocol (used only in the complexity-of-maintenance discussion). When discussing a particular identification/verification task, we may write s for the number of attributes actually queried/traversed by the procedure (e.g., interface or schema checks in software/data systems, phenotypic checks in taxonomy), with $s \leq n$. The maintenance-only parameter m appears only in Section III.

Definition I.5 (Attribute profile)**.** The attribute profile function $\pi : \mathcal{V} \to \{0,1\}^{|\mathcal{I}|}$ maps each value to its complete attribute signature:

$$\pi(v) = (q_I(v))_{I \in \mathcal{I}}$$

Definition I.6 (Attribute indistinguishability)**.** Values $v, w \in \mathcal{V}$ are *attribute-indistinguishable*, written $v \sim w$, iff $\pi(v) = \pi(w)$.

The relation $\sim$ is an equivalence relation. We write $[v]_\sim$ for the equivalence class of v.

Definition I.7 (Attribute-only observer)**.** An *attribute-only observer* is any procedure whose interaction with a value $v \in \mathcal{V}$ is limited to queries in $\Phi_{\mathcal{I}}$. Formally, the observer interacts with v only via primitive attribute queries $q_I \in \Phi_{\mathcal{I}}$; hence any transcript (and output) factors through $\pi(v)$.

C. The Central Question

The central question is: **what semantic properties can an attribute-only observer compute?**

A semantic property is a function $P : \mathcal{V} \to \{0,1\}$ (or more generally, $P : \mathcal{V} \to Y$ for some codomain Y). We say P is *attribute-computable* if there exists a function $f : \{0,1\}^{|\mathcal{I}|} \to Y$ such that $P(v) = f(\pi(v))$ for all v.

D. The Information Barrier

Theorem I.8 (Information barrier)**.** *Let $P : \mathcal{V} \to Y$ be any function. If P is attribute-computable, then P is constant on $\sim$-equivalence classes:*

$$v \sim w \implies P(v) = P(w)$$

Equivalently: no attribute-only observer can compute any property that varies within an equivalence class.

(L: ACS8, ACS9, COH1)

Proof. Suppose P is attribute-computable via f, i.e., $P(v) = f(\pi(v))$ for all v. Let $v \sim w$, so $\pi(v) = \pi(w)$. Then:

$$P(v) = f(\pi(v)) = f(\pi(w)) = P(w)$$

■

Remark I.9 (Information-theoretic nature). The barrier is *informational*, not computational. Given unlimited time, memory, and computational power, an attribute-only observer still cannot distinguish v from w when $\pi(v) = \pi(w)$. The constraint is on the evidence itself.

Remark I.10 (Role in the paper). Theorem I.8 is the base invariance statement. The technical contribution is the downstream structure built on top of it: the ambiguity-based converse (Theorem II.36), the Pareto characterization (Theorem VI.3), and the matroid/equicardinality results (Section IV).

Corollary I.11 (Class identity is not attribute-computable)**.** *Let $C : \mathcal{V} \to \{1, \ldots, k\}$ be the class assignment function. If there exist values v, w with $\pi(v) = \pi(w)$ but $C(v) \neq C(w)$, then class identity is not attribute-computable.*

(L: ACS9)

Proof. Direct application of Theorem I.8 to $P = C$. ■

E. The Positive Result: Nominal Tagging

We now show that augmenting attribute observations with a single primitive, nominal-tag access, achieves constant witness cost.

Definition I.12 (Nominal-tag access)**.** A *nominal tag* is a value $\tau(v) \in \mathcal{T}$ associated with each $v \in \mathcal{V}$, representing the class identity of v. The *nominal-tag access* operation returns $\tau(v)$ in $O(1)$ time.

Definition I.13 (Primitive query set)**.** The extended primitive query set is $\Phi_{\mathcal{I}}^+ = \Phi_{\mathcal{I}} \cup \{\tau\}$, where τ denotes nominal-tag access.

Definition I.14 (Witness cost)**.** Let $W(P)$ denote the minimum number of primitive queries from $\Phi_{\mathcal{I}}^+$ required to compute property P. We distinguish two tasks:

- W_{id}: Cost to identify the class of a single entity.
- W_{eq}: Cost to determine if two entities have the same class.

Unless specified, W refers to W_{eq}.

Theorem I.15 (Constant witness for class identity)**.** *Under nominal-tag access, class identity checking has constant witness cost:*

$$W(\textit{class-identity}) = O(1)$$

Specifically, the witness procedure is: return $\tau(v_1) = \tau(v_2)$*.*

(L: ACS7, FORC2, DOM1)

Proof. The procedure makes exactly 2 primitive queries (one τ access per value) and one comparison. This is $O(1)$ regardless of the number of available attributes $|\mathcal{I}|$. ■

Theorem I.16 (Attribute-only lower bound)**.** *For attribute-only observers, class identity checking requires:*

$$W(\textit{class-identity}) = \Omega(d)$$

in the worst case, where d is the distinguishing dimension (Definition IV.9).

(L: ACS1)

Proof. Assume a zero-error attribute-only procedure halts after fewer than d queries on every execution path. Fix any execution path and let $Q \subseteq \mathcal{I}$ be the set of queried attributes on that path, so $|Q| < d$. Since d is the cardinality of every minimal distinguishing set, no set of size $< d$ is distinguishing; hence there exist values v, w from different classes with identical answers on all attributes in Q.

An adversary can answer the procedure's queries consistently with both v and w along this path. Therefore the resulting transcript (and output) is identical on v and w, contradicting zero-error class identification. So some execution path must use at least d queries, giving worst-case cost $\Omega(d)$. ■

F. Main Contributions

This paper establishes the following results:

1) **Information Barrier Theorem** (Theorem I.8): Attribute-only observers cannot compute any property that varies within $\sim$-equivalence classes. This is an observational impossibility, not a computational limitation, and it is the source of the downstream rate-query tradeoffs.
2) **Constant-Witness Theorem** (Theorem I.15): Nominal-tag access achieves constant witness cost for class identity. Attribute-only observers have matching lower bound $\Omega(d)$ (Theorem I.16), where d is the distinguishing dimension (Definition IV.9).
3) **Complexity Separation** (Section III): We establish O(1) vs O(k) vs $\Omega(d)$ complexity bounds for error localization under different observation regimes (where d is the distinguishing dimension).
4) **Matroid Structure** (Section IV): Minimal distinguishing query sets form the bases of a matroid. All such sets have equal cardinality, so the distinguishing dimension is an invariant of the observation model rather than an artifact of a particular proof or query order.
5) (L, W, D) **Optimality** (Section VI): We characterize the zero-error converse via collision multiplicity A_π and prove uniqueness of the nominal point in the maximal-barrier regime ($A_\pi = k$), turning an informal heuristic into an explicit resource law.
6) **Finite-Block Confusability Law** (Section II): We prove exact block thresholds for zero-error feasibility in the confusability formulation: at blocklength t, feasibility is equivalent to an alphabet threshold A_π^t, with exact linear log-bit scaling and per-coordinate stabilization.
7) **Open-World Robustness + Decidability Boundary**: We formalize that barrier-freedom is not extension-stable in open worlds (any finite barrier-free world can be extended to force

a collision), and we prove a Rice-style impossibility theorem: no computable certifier can decide barrier existence/freedom for arbitrary generators.

8) **Machine-Checked Proofs**: All results formalized in Lean 4 (8092 lines, 352 theorem/lemma statements, 0 `sorry` placeholders).

(L: ROB1, ROB2, UND1, UND2)

Without the equicardinality result behind the matroid structure, the lower bound would depend on which distinguishing set or query strategy one chose. The matroid theorem is what makes d strategy-independent.

G. Related Work and Positioning

This paper does not develop asymptotic coding theorems or claim new zero-error capacity results. Its contribution is to make explicit the bit/query/error tradeoffs governing zero-error classification under constrained observations.

Identification via channels. Our work is adjacent to the identification paradigm introduced by Ahlswede and Dueck [1], [2]. In their framework, a decoder need not reconstruct a message but only answer "is the message m?" for a given hypothesis. Our setting is different in three ways: (1) we consider one-shot zero-error class identification rather than asymptotic vanishing-error identification, (2) queries are adaptive rather than block codes, and (3) we allow auxiliary tagging (rate L) to reduce query cost. The connection is motivational and terminological rather than a direct reuse of the original coding theorem.

Rate-distortion theory. The (L, W, D) framework is analogous to rate-distortion theory [3], [4]: the "distortion" D is semantic (class misidentification), and there is a second resource W (query cost) alongside rate L. Classical rate-distortion asks for the minimum rate needed to achieve distortion D; here the main question is which combinations of tag bits and queries permit $D = 0$. Theorem VI.3 gives the resulting converse in terms of collision multiplicity and identifies the unique nominal point in the maximal-barrier regime.

Rate-distortion-perception tradeoffs. Blau and Michaeli [5] extended rate-distortion theory by adding a perception constraint, creating a three-way tradeoff. Our query cost W plays an analogous role: it measures the interactive cost of achieving low distortion rather than a distributional constraint. We cite this as a structural analogy, not as a claim that the present model inherits the full geometry of that setting.

Zero-error information theory. The matroid structure (Section IV) places the model near zero-error information theory. Körner [6] and Witsenhausen [7] studied zero-error source coding where confusable symbols must be distinguished. Here the direct theorem is simpler: when $L = 0$, the distinguishing dimension (Definition IV.9) is the minimum number of binary queries needed for zero-error class separation. In addition, the paper proves exact finite-block confusability laws: block feasibility threshold A_π^k, exact linear log-bit scaling, and per-coordinate stabilization in the worst-case regime. These are formalized in Lean and strengthen the operational converse while remaining distinct from full distribution-optimized graph-entropy/capacity theorems.

Query complexity and communication complexity. The $\Omega(d)$ lower bound for attribute-only identification relates to decision tree complexity [8] and interactive communication [9]. The key distinction is that our queries are constrained to a fixed observable family $\mathcal{I}$, not arbitrary predicates.

Compression in classification systems. The same ambiguity bound appears across databases, knowledge graphs, taxonomy, and software-runtime typing systems: for zero-error identification, ambiguity induces a minimum metadata requirement $L \geq \log_2 A_\pi$ (Theorem II.36), with maximal-barrier specialization $L \geq \log_2 k$.

Runtime/schema instantiation (secondary). In nominal-vs-structural interface settings [10], [11], the model yields a concrete cost statement: under attribute collisions, purely profile-based identification has worst-case $\Omega(d)$ witness cost, while explicit tags achieve $O(1)$ identification using $O(\log A_\pi)$ bits. This is one instantiation of the general bounds.

H. Paper Organization

Section II formalizes the compression framework and defines the (L, W, D) tradeoff. Section III establishes complexity bounds for error localization. Section IV proves the matroid structure of distinguishing query families. Section V analyzes witness cost in detail. Section VI proves the ambiguity-based converse and Pareto characterization. Section VIII concludes. Extended instantiations and the Lean formalization are provided in the supplementary material.

II. Compression Framework

A. Semantic Compression: The Problem

The fundamental problem of *semantic compression* is: given a value v from a large space $\mathcal{V}$, how can we represent v compactly while preserving the ability to answer semantic queries about v? This differs from classical source coding in that the goal is not reconstruction but *identification*: determining which equivalence class v belongs to.

Classical rate-distortion theory [3] studies the tradeoff between representation size and reconstruction fidelity. We extend this to a discrete classification setting with three dimensions: *tag length* L (bits of storage), *witness cost* W (queries or bits of communication required to determine class membership), and *distortion* D (misclassification probability).

This work treats contemporary classification problems (databases, knowledge graphs, model registries) with explicit resource accounting. The point is not to claim a new Shannon-style asymptotic coding theorem, but to make explicit the bit/query/error tradeoffs that already constrain zero-error identification in practice.

B. Universe of Discourse

Definition II.1 (Classification scheme)**.** A *classification scheme* is any procedure (deterministic or randomized), with arbitrary time and memory, whose only access to a value $v \in \mathcal{V}$ is via:

1) The *observation family* $\Phi = \{q_I : I \in \mathcal{I}\}$, where $q_I(v) = 1$ iff v satisfies attribute I; and optionally
2) A *nominal-tag primitive* $\tau : \mathcal{V} \to \mathcal{T}$ returning an opaque class identifier.

All theorems in this paper quantify over all such schemes.

This definition is intentionally broad: schemes may be adaptive, randomized, or computationally unbounded. The constraint is *observational*, not computational.

Theorem II.2 (Information barrier)**.** *For all classification schemes with access only to Φ (no nominal tag), the output is constant on $\sim_\Phi$-equivalence classes. Therefore, no such scheme can compute any property that varies within a $\sim_\Phi$-class.*

(L: ACS8, ACS9, COH1)

Proof. Let $v \sim_\Phi w$, meaning $q_I(v) = q_I(w)$ for all $I \in \mathcal{I}$. Any scheme's execution trace depends only on query responses. Since all queries return identical values for v and w, the scheme cannot distinguish them. Any output must therefore be identical. ■

Proposition II.3 (Model capture)**.** *Any real-world classification protocol whose evidence consists solely of attribute-membership queries is representable as a scheme in the above model. Conversely, any additional capability corresponds to adding new observations to Φ.*

(L: ACS5, COMPL1, FORC2)

This proposition forces any objection into a precise form: to claim the theorem does not apply, one must name the additional observation capability not in Φ. "Different universe" is not a coherent objection; it must reduce to "I have access to oracle $X \notin \Phi$."

C. Two-Axis Instantiation (Representative Domain Example)

The core results require only $(\mathcal{V}, C, \pi)$ and the observation family Φ. The two-axis decomposition below is one representative instantiation and is not an additional axiom for the general theorems.

In that instantiation, each value is characterized by:

- **Provenance axis** (B): source/lineage information that can distinguish entities with identical observable profiles[1]
- **Profile axis** (S): the observable attribute profile induced by Φ

[1]In the Lean witness model, this axis is represented as `Bases`.

Definition II.4 (Two-axis representation)**.** A value $v \in \mathcal{V}$ has representation $(B(v), S(v))$ where:

$$B(v) = \text{provenance}(v) \tag{1}$$

$$S(v) = \pi(v) = (q_I(v))_{I \in \mathcal{I}} \tag{2}$$

The provenance axis captures *identity-bearing origin information*; the profile axis captures *observable behavior/attributes*.

In concrete deployments, B may be carried by lineage metadata, registry identifiers, or equivalent provenance carriers. Any implementation-specific normalization or lookup machinery is auxiliary.

Theorem II.5 (Fixed-axis completeness)**.** *Let a fixed-axis domain be specified by an axis map $\alpha : \mathcal{V} \to \mathcal{A}$ and an observation family Φ such that each primitive query $q \in \Phi$ factors through α. Then every in-scope semantic property (i.e., any property computable by an admissible Φ-only strategy) factors through α: there exists $\tilde{P}$ with*

$$P(v) = \tilde{P}(\alpha(v)) \quad \textit{for all } v \in \mathcal{V}.$$

(L: ACS5, COMPL1, DOM1)

In this two-axis instantiation, $\alpha(v) = (B(v), S(v))$, so in-scope semantic properties are functions of (B, S).

Proof. An admissible Φ-only strategy observes v solely through responses to primitive queries $q_I \in \Phi$. By hypothesis each such response is a function of $\alpha(v)$. Therefore every query transcript, and hence any strategy's output, depends only on $\alpha(v)$, so the computed property factors through α. ■

Corollary II.6 (Fixed-Axis Incompleteness)**.** *Any fixed-axis classification system is complete only for properties measurable on the fixed axis map α, and incomplete for any property that varies within an α-fiber. Equivalently, if $\alpha(v) = \alpha(w)$ but $P(v) \neq P(w)$, then no admissible Φ-only strategy can compute P with zero error.*

(L: L2)

D. Attribute Equivalence and Observational Limits

Recall from Section 1 the attribute equivalence relation:

Definition II.7 (Attribute equivalence (restated))**.** Values $v, w \in \mathcal{V}$ are attribute-equivalent, written $v \sim w$, iff $\pi(v) = \pi(w)$, i.e., they induce exactly the same attribute responses.

Proposition II.8 (Equivalence class structure)**.** *The relation $\sim$ partitions $\mathcal{V}$ into equivalence classes. Let $\mathcal{V}/{\sim}$ denote the quotient space. An attribute-only observer effectively operates on $\mathcal{V}/{\sim}$, not $\mathcal{V}$.*

(L: ACS8, COMPL1, FORC2)

Corollary II.9 (Information loss quantification)**.** *The information lost by attribute-only observation is:*

$$H(\mathcal{V}) - H(\mathcal{V}/{\sim}) = H(\mathcal{V}|\pi)$$

where H denotes entropy. This quantity is positive whenever multiple classes share the same attribute profile.

(L: ACS8)

E. One-Shot Zero-Error Feasibility

We now formalize the identification problem in channel-theoretic terms. Let $C : \mathcal{V} \to \{1, \ldots, k\}$ denote the class assignment function, and let $\pi : \mathcal{V} \to \{0, 1\}^n$ denote the attribute profile.

Definition II.10 (Observation channel)**.** The *observation channel* induced by observation family Φ is the mapping $C \to \pi(V)$ for a random entity V drawn from distribution P_V over $\mathcal{V}$. The channel output is the attribute profile; the channel input is implicitly the class $C(V)$.

Theorem II.11 (One-Shot Zero-Error Identification Feasibility)**.** *Let $\mathcal{C} = \{1, \ldots, k\}$ be the class space. The* zero-error identification capacity *of the observation channel is:*

$$C_{id} = \begin{cases} \log_2 k & \text{if } \pi \text{ is injective on classes} \\ 0 & \text{otherwise} \end{cases}$$

That is, zero-error identification of all k classes is achievable if and only if every class has a distinct attribute profile. When π is not class-injective, no rate of identification is achievable with zero error.

(L: ACS8)

Proof. Achievability: If π is injective on classes, then observing $\pi(v)$ determines $C(v)$ uniquely. The decoder simply inverts the class-to-profile mapping.

Converse (deterministic): Suppose two distinct classes $c_1 \neq c_2$ share a profile: $\exists v_1 \in c_1, v_2 \in c_2$ with $\pi(v_1) = \pi(v_2)$. Then any decoder $g(\pi(v))$ outputs the same class label on both v_1 and v_2, so it cannot be correct for both. Hence zero-error identification of all classes is impossible. ■

Remark II.12 (Information-theoretic corollary)*.* Under any distribution with positive mass on both colliding classes, $I(C; \pi(V)) < H(C)$. This is an average-case consequence of the deterministic barrier above.

Remark II.13 (Relation to Ahlswede-Dueck)*.* In the identification paradigm of [1], the decoder asks "is the message m?" rather than "what is the message?" This yields double-exponential codebook sizes. Our setting is different: we require zero-error identification of the *class*, not hypothesis testing. The one-shot zero-error identification feasibility threshold (π must be class-injective) is binary rather than a rate.

Remark II.14 (Terminology)*.* Theorem II.11 is a one-shot feasibility statement, not a Shannon asymptotic coding theorem. The notation is retained only as a compact summary of the zero-error/non-zero-error dichotomy under the stated observation model.

The key insight is that tagging provides a *side channel* that restores identifiability when the attribute channel fails:

Theorem II.15 (Tag-Restored Zero Error (Sufficiency))**.** *A class-injective nominal tag of length $L \geq \lceil \log_2 k \rceil$ bits suffices for zero-error identification, regardless of whether π is class-injective.*

(L: ACS7)

Proof. A nominal tag $\tau : \mathcal{V} \to \{1, \dots, k\}$ assigns a unique identifier to each class. Reading $\tau(v)$ determines $C(v)$ in $O(1)$ queries, independent of the attribute channel. ■

F. Witness Cost: Query Complexity for Semantic Properties

Definition II.16 (Witness procedure)**.** A *witness procedure* for property $P : \mathcal{V} \to Y$ is an algorithm A that:

1) Takes as input a value $v \in \mathcal{V}$ (via query access only)
2) Makes queries to the primitive set $\Phi^+_{\mathcal{I}}$
3) Outputs $P(v)$

Definition II.17 (Witness cost)**.** The *witness cost* of property P is:

$$W(P) = \min_{A \text{ computes } P} c(A)$$

where $c(A)$ is the worst-case number of primitive queries made by A.

Remark II.18 (Relationship to query complexity). Witness cost is a form of query complexity [8] specialized to semantic properties. Unlike Kolmogorov complexity, W is computable and depends on the primitive set, not a universal machine.

Lemma II.19 (Witness cost lower bounds)**.** *For any property P:*

1) *If P is attribute-computable: $W(P) \leq |\mathcal{I}|$*
2) *If P varies within some $\sim$-class: $W(P) = \infty$ for attribute-only observers*
3) *With nominal-tag access: $W(\text{class-identity}) = O(1)$*

(L: ACS8, ACS7, ACS1)

G. *The* (L, W, D) *Tradeoff*

We now define the three-dimensional tradeoff space that characterizes observation strategies, using information-theoretic units.

Definition II.20 (Tag rate)**.** For a set of class identifiers (tags) $\mathcal{T}$ with $|\mathcal{T}| = k$, the *tag rate* L is the minimum number of bits required to encode a class identifier:

$$L \geq \log_2 k \quad \text{bits per value}$$

For nominal-tag observers, $L = \lceil \log_2 k \rceil$ (optimal prefix-free encoding). For attribute-only observers, $L = 0$ (no explicit tag stored). Under a distribution P over classes, the expected tag length is $\mathbb{E}[L] \geq H(P)$ by Shannon's source coding theorem [3].

Definition II.21 (Witness cost (Query/Communication complexity))**.** The *witness cost* W is the minimum number of primitive queries (or bits of interactive communication) required for class identification:

$$W = \min_{A \text{ decides class}} c(A)$$

where $c(A)$ is the worst-case query count. This is a form of query complexity [8] or interactive identification cost.

Definition II.22 (Class estimator)**.** Fix class map $C : \mathcal{V} \to \{1, \dots, k\}$. An observation strategy g induces an estimate

$$\hat{C}_g(v; \omega) \in \{1, \dots, k\}$$

from the available evidence (tag bits, query transcript, and optional internal randomness ω).

Definition II.23 (Distortion indicator and expected distortion)**.** For strategy g, define

$$d_g(v; \omega) := \mathbf{1}\left[\hat{C}_g(v; \omega) \neq C(v)\right].$$

Under data distribution P_V and strategy randomness, expected distortion is

$$D(g) = \Pr_{v \sim P_V, \omega} \big[\hat{C}_g(v; \omega) \neq C(v)\big].$$

The zero-error regime is $D(g) = 0$.

Remark II.24 (Interpretation). In this paper, D is strictly class-misidentification probability. Additional semantic notions (e.g., hierarchical or behavior-weighted penalties) are treated as extensions in the supplementary material.

H. The (L, W, D) Tradeoff Space

Admissible schemes. To make the Pareto-optimality claim precise, we specify the class of admissible observation strategies:

- **Deterministic or randomized**: Schemes may use randomness; W is worst-case query count.
- **Computationally unbounded**: No time/space restrictions; the constraint is observational.
- **No preprocessing over class universe**: The scheme cannot precompute a global lookup table indexed by all possible classes.
- **Tags are injective on classes**: A nominal tag $\tau(v)$ uniquely identifies the class of v. Variable-length or compressed tags are permitted; L counts bits.
- **No amortization across queries**: W is per-identification cost, not amortized over a sequence.

Justification. The "no preprocessing" and "no amortization" constraints exclude trivializations:

- *Preprocessing*: With unbounded preprocessing over the class universe $\mathcal{T}$, one could build a lookup table mapping attribute profiles to classes. This reduces identification to $O(1)$ table lookup, but the table has size $O(|\mathcal{T}|)$, hiding the complexity in space rather than eliminating it. The constraint models systems that cannot afford $O(|\mathcal{T}|)$ storage per observer.
- *Amortization*: If W were amortized over a sequence of identifications, one could cache earlier results. This again hides complexity in state. The per-identification model captures stateless observers (typical in database queries, taxonomy lookup, and protocol/classification services).

Dropping these constraints changes the achievable region but not the qualitative separation: nominal tags still dominate for $D = 0$ because they provide $O(1)$ worst-case identification without requiring $O(|\mathcal{T}|)$ preprocessing.

Under these rules, "dominance" means strict improvement on at least one of (L, W, D) with no regression on others.

Definition II.25 (Achievable region)**.** A point (L, W, D) is *achievable* if there exists an admissible observation strategy realizing those values. Let $\mathcal{R} \subseteq \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times [0, 1]$ denote the achievable region.

Definition II.26 (Pareto optimality)**.** A point (L^*, W^*, D^*) is *Pareto-optimal* if there is no achievable (L, W, D) with $L \leq L^*$, $W \leq W^*$, $D \leq D^*$, and at least one strict inequality.

The main result of Section VI is: (i) a converse in terms of collision multiplicity A_π, and (ii) uniqueness of the nominal $D = 0$ Pareto point in the maximal-barrier regime.

Definition II.27 (Information-barrier domain)**.** A classification domain has an *information barrier* (relative to Φ) if there exist distinct classes $c_1 \neq c_2$ with identical Φ-profiles. Equivalently, π is not injective on classes.

Definition II.28 (Collision multiplicity)**.** Let $\mathcal{C} = \{1, \ldots, k\}$ and let $\pi_{\mathcal{C}} : \mathcal{C} \to \{0, 1\}^n$ be the class-level profile map. Define

$$A_\pi := \max_{u \in \mathrm{Im}(\pi_{\mathcal{C}})} \left| \{c \in \mathcal{C} : \pi_{\mathcal{C}}(c) = u\} \right| .$$

Thus A_π is the size of the largest class-collision block under observable profiles.

Definition II.29 (Maximal-Barrier Regime)**.** The domain is *maximal-barrier* if $A_\pi = k$, i.e., all classes collide under the observation map.

I. Open-World Robustness and Decidability Boundary

Definition II.30 (Finite realized world)**.** Let $\mathcal{C}$ be the class universe. A *finite realized world* is a finite subset $W \subseteq \mathcal{C}$ of classes currently present in a deployed system snapshot. We call W

barrier-free iff

$$\forall c_1, c_2 \in W,\ \pi_{\mathcal{C}}(c_1) = \pi_{\mathcal{C}}(c_2) \Rightarrow c_1 = c_2.$$

Definition II.31 (Open-world extension)**.** An *open-world extension* is a super-world $W' \supseteq W$ obtained by adding classes while keeping the observation family Φ fixed.

Theorem II.32 (Barrier-freedom is not extension-stable)**.** *In the open-world class model formalized in Lean, it is not true that barrier-freedom is preserved under all extensions:*

$$\neg\Big(\forall W \subseteq W',\ \textit{BarrierFree}(W) \Rightarrow \textit{BarrierFree}(W')\Big).$$

(L: ROB1, ROB2)

Proof. Take the empty world $W_0 = \varnothing$, which is barrier-free. In the Lean model, two concrete classes (`ConfigType` and `StepConfigType`) are shape-equivalent (same observable profile) but nominally distinct (different lineage), so adding them yields an extension $W_1 \supseteq W_0$ that is not barrier-free. Therefore barrier-freedom is not extension-stable. ∎

Definition II.33 (Function-level barrier predicate)**.** For a partial observer generator $f : \mathbb{N} \rightharpoonup \mathbb{N}$, define

$$\mathrm{HasBarrier}(f)\ :\Leftrightarrow\ \exists x \neq y, \exists z,\ z \in f(x) \wedge z \in f(y).$$

Define $\mathrm{BarrierFree}(f) :\Leftrightarrow \neg\mathrm{HasBarrier}(f)$.

Theorem II.34 (Rice-style non-computability of barrier certification)**.** *There is no computable predicate on program codes deciding whether the generated observer has a barrier:*

$$\neg\mathrm{ComputablePred}(c \mapsto \mathrm{HasBarrier}(\mathrm{eval}(c)))\,.$$

Equivalently, barrier-freedom certification is also non-computable:

$$\neg\mathrm{ComputablePred}(c \mapsto \mathrm{BarrierFree}(\mathrm{eval}(c)))\,.$$

(L: UND1, UND2)

Proof. Assume there is a computable certifier for $P(c) := \mathrm{HasBarrier}(\mathrm{eval}(c))$. Set

$$\mathcal{C}_{\mathrm{bar}} := \{f : \mathbb{N} \rightharpoonup \mathbb{N} \mid \mathrm{HasBarrier}(f)\}.$$

Then $P(c)$ is exactly membership of $\mathrm{eval}(c)$ in $\mathcal{C}_{\mathrm{bar}}$. The property is non-trivial:

$$\mathrm{HasBarrier}(\lambda x.\, 0) \quad \text{and} \quad \neg\mathrm{HasBarrier}(\lambda x.\, x+1).$$

Rice's theorem (formalized in Lean and instantiated with these two witnesses) implies that if membership in $\mathcal{C}_{\mathrm{bar}}$ were computable from code, then every partial recursive function would have to lie in $\mathcal{C}_{\mathrm{bar}}$, contradicting the successor witness. Hence P is not computable. The barrier-free statement is equivalent by

$$\mathrm{BarrierFree}(f) \iff \neg\mathrm{HasBarrier}(f),$$

so computability of one would imply computability of the other. ■

Corollary II.35 (Certification consequence for persistent $D = 0$ claims)**.** *For unrestricted open-world generators, one cannot computably certify that attribute-only identification remains barrier-free under all future extensions. Consequently, persistent $D = 0$ guarantees cannot rely on "currently collision-free" attribute structure alone.*

J. Converse: Tag Rate Lower Bound

Theorem II.36 (Converse)**.** *For any classification domain, any scheme achieving $D = 0$ requires*

$$2^L \geq A_\pi \quad \textit{equivalently} \quad L \geq \log_2 A_\pi,$$

where A_π is the collision multiplicity from Definition II.28.

(L: LWDC1)

Proof. Fix a collision block $G \subseteq \mathcal{C}$ with $|G| = A_\pi$ and identical observable profile. For classes in G, query transcripts are identical, so zero-error decoding must separate those classes using tag outcomes. With L tag bits there are at most 2^L outcomes, hence $2^L \geq |G| = A_\pi$. ■

Corollary II.37 (Maximal-Barrier Converse)**.** *If the domain is maximal-barrier ($A_\pi = k$), any zero-error scheme satisfies $L \geq \log_2 k$.*

(L: LWDC3)

K. Confusability Graph Thresholds and Block Scaling

Let $\pi_{\mathcal{C}} : \mathcal{C} \to \{0,1\}^n$ be the class-profile map from Definition II.28. The induced confusability graph connects two distinct classes when they share the same observable profile.

Theorem II.38 (Exact one-shot feasibility threshold)**.** *For an alphabet of n nominal tags, zero-error class tagging is feasible if and only if*

$$A_\pi \leq n.$$

(L: GPH13)

Proof. Classes in the same profile fiber are pairwise confusable, so each such fiber is a clique in the confusability graph. Zero-error tagging is equivalent to a proper graph coloring; therefore feasibility is equivalent to having at least as many colors as the largest clique size, which is exactly A_π. ■

Theorem II.39 (Exact block feasibility law)**.** *For block length $t \geq 1$ with coordinatewise observation on $\mathcal{C}^t$, zero-error tagging with alphabet size n_t is feasible if and only if*

$$A_\pi^t \leq n_t.$$

Equivalently, the minimal feasible block alphabet is exactly A_π^t.

(L: GPH16, GPH17, GPH20)

Corollary II.40 (Linear log-bit scaling)**.** *Let $L_t^\star := \log_2(A_\pi^t)$ be the minimum block tag budget in bits. Then*

$$L_t^\star = t \log_2 A_\pi, \qquad \frac{L_t^\star}{t} = \log_2 A_\pi.$$

TABLE I
IDENTIFICATION STRATEGIES FOR 1000 CLASSES WITH 50 ATTRIBUTES.

Strategy	**Tag** L	**Witness** W
Explicit-tag identification	$\lceil \log_2 1000 \rceil = 10$ bits	$O(1)$
Attribute-only (exhaustive)	0	≤ 50 queries
Attribute-only (adaptive)	0	$\geq d$ queries

So per-entity tag rate is exactly stable across block length.

(L: GPH18, GPH19, GPH24)

Remark II.41 (Scope). These are worst-case finite-block confusability laws. They strengthen the zero-error converse with exact block thresholds, while remaining distinct from full distribution-optimized Körner graph-entropy theory.

L. Lossy Regime: Deterministic vs Noisy Models

The zero-error corner ($D = 0$) is governed by Theorem II.36. For $D > 0$, the model matters:

- **Deterministic queries (this section):** there is no universal law of the form $W = O(\log(1/\epsilon) \cdot d)$. If classes collide on all deterministic observations, repeating those same observations does not reduce error.
- **Noisy queries:** repeated independent observations can reduce error exponentially, yielding logarithmic-in-$1/\epsilon$ sample complexity.

Thus, in the deterministic model, distortion is controlled by collision geometry and decision rules; in noisy models, repetition-based concentration bounds become relevant.

M. Concrete Example

Consider a classification system with $k = 1000$ classes, each characterized by a subset of $n = 50$ binary attributes. Table I compares the strategies.

Here d is the distinguishing dimension, the size of any minimal distinguishing query set. For many practical systems, d is well below n but nonzero. The gap between 10 bits of storage versus 5–50 queries per identification is the cost of forgoing explicit tags.

III. Complexity Bounds

A. The Error Localization Theorem

a) Scope of this section.: This section studies *maintenance/localization* complexity, measured by locations to inspect ($E(\cdot)$), not per-instance identification complexity (W) from Sections II and V. The metrics are related but distinct: W quantifies online class-identification effort, while E quantifies where constraint logic is distributed in implementations.

Definition III.1 (Error location count)**.** Let $E(\mathcal{O})$ be the number of locations that must be inspected to find all potential violations of a constraint under observation family $\mathcal{O}$.

Theorem III.2 (Nominal-tag localization)**.** $E(\textit{nominal-tag}) = O(1)$.

(L: ACS7, DOM1, FORC2)

Proof. Under nominal-tag observation, the constraint "v must be of class A" is satisfied iff $\tau(v) \in \text{subtypes}(A)$. This is determined at a single location: the definition of $\tau(v)$'s class. One location. ∎

Theorem III.3 (Declared-attribute localization)**.** *$E(\textit{attribute-only, declared}) = O(k)$ where k = number of entity classes.*

(L: ACS3)

Proof. With declared attribute sets, the constraint "v must satisfy attribute I" requires verifying that each class satisfies all attributes in I. For k classes, $O(k)$ locations. ∎

Theorem III.4 (Attribute-only localization)**.** *$E(\textit{attribute-only}) = \Omega(m)$ where m = number of query sites.*

(L: ACS3, ACS9, COH1)

Proof. Under attribute-only observation, each query site independently checks "does v have attribute a?" with no centralized declaration. For m query sites, each must be inspected. Lower bound is $\Omega(m)$. ∎

Corollary III.5 (Strict dominance)**.** *Nominal-tag observation strictly dominates attribute-only:* $E(\textit{nominal-tag}) = O(1) < \Omega(m) = E(\textit{attribute-only})$ *for all* $m > 1$.

(L: ACS4, DOM1, FORC2)

B. The Information Scattering Theorem

Definition III.6 (Constraint encoding locations)**.** Let $I(\mathcal{O}, c)$ be the set of locations where constraint c is encoded under observation family $\mathcal{O}$.

Theorem III.7 (Attribute-only scattering)**.** *For attribute-only observation,* $|I(\textit{attribute-only}, c)| = O(m)$ *where* m *= query sites using constraint* c*.*

(L: ACS3)

Proof. Each attribute query independently encodes the constraint. No shared reference exists. Constraint encodings scale with query sites. ∎

Theorem III.8 (Nominal-tag centralization)**.** *For nominal-tag observation,* $|I(\textit{nominal-tag}, c)| = O(1)$.

(L: ACS6, UNIQ1, DOM1)

Proof. The constraint "must be of class A" is encoded once in the definition of A. All tag checks reference this single definition. ∎

Corollary III.9 (Maintenance entropy)**.** *Attribute-only observation maximizes maintenance entropy; nominal-tag observation minimizes it.*

(L: ACS2, COH1, FORC2)

IV. Matroid Structure

A. Model Contract (Fixed-Axis Domains)

Model contract (fixed-axis domain). A domain is specified by a fixed observation family Φ derived from a fixed axis map $\alpha : \mathcal{V} \to \mathcal{A}$ (e.g., $\alpha(v) = (B(v), S(v))$). An observer is permitted to interact with v only through primitive queries in Φ, and each primitive query factors through α: for every $q \in \Phi$, there exists $\tilde{q}$ such that $q(v) = \tilde{q}(\alpha(v))$. A property is in-scope semantic iff it is computable by an admissible strategy that uses only responses to queries in Φ (under our admissibility constraints: no global preprocessing tables, no amortized caching, etc.).

We adopt Φ as the complete observation universe for this paper: to claim applicability to a concrete runtime one must either (i) exhibit mappings from each runtime observable into Φ, or (ii) enforce the admissibility constraints (no external registries, no reflection, no preprocessing/amortization). Under either condition the theorems apply without qualification.

Proposition IV.1 (Observational Quotient)**.** *For any admissible strategy using only Φ, the entire interaction transcript (and hence the output) depends only on $\alpha(v)$. Equivalently, any in-scope semantic property P factors through α: there exists $\tilde{P}$ with $P(v) = \tilde{P}(\alpha(v))$ for all v.*

(L: ACS8, ACS9, COH1)

Corollary IV.2 (Why "ad hoc" = adding an axis/tag)**.** *If two values v, w satisfy $\alpha(v) = \alpha(w)$, then no admissible Φ-only strategy can distinguish them with zero error. Any mechanism that does distinguish such pairs must introduce additional information not present in α (equivalently, refine the axis map by adding a new axis/tag).*

(L: ACS8, COMPL1, FORC2)

B. Query Families and Distinguishing Sets

The classification problem is: given a set of queries, which subsets suffice to distinguish all entities?

Definition IV.3 (Query family)**.** Let $\mathcal{Q}$ be the set of all primitive queries available to an observer. For a classification system with attribute set $\mathcal{I}$, we have $\mathcal{Q} = \{q_I : I \in \mathcal{I}\}$ where $q_I(v) = 1$ iff v satisfies attribute I.

In this section, "queries" are the primitive attribute predicates $q \in \Phi$ (equivalently, each q factors through the axis map: $q = \tilde{q} \circ \alpha$). See the Convention above where $\Phi := \mathcal{Q}$.

Convention: $\Phi := \mathcal{Q}$. All universal quantification over "queries" ranges over $q \in \Phi$ only.

Definition IV.4 (Distinguishing set)**.** A subset $S \subseteq \mathcal{Q}$ is *distinguishing* if, for all values v, w with $\mathrm{class}(v) \neq \mathrm{class}(w)$, there exists $q \in S$ such that $q(v) \neq q(w)$.

Definition IV.5 (Minimal distinguishing set)**.** A distinguishing set S is *minimal* if no proper subset of S is distinguishing.

C. Matroid Structure of Query Families

Scope and assumptions. The matroid theorem below is unconditional within the fixed-axis observational theory defined above. In this section, "query" always means a primitive predicate $q \in \Phi$ (equivalently, q factors through α as in the Model Contract). It depends only on:

- $E = \Phi$ is the ground set of primitive queries (attribute predicates).
- "Distinguishing": for all values v, w with $\mathrm{class}(v) \neq \mathrm{class}(w)$, there exists $q \in S$ such that $q(v) \neq q(w)$ (Def. above).
- "Minimal" means inclusion-minimal: no proper subset suffices.

No further assumptions are required within this theory beyond the fixed observation family Φ. In Lean, the mechanization is explicit end-to-end. Closure-operator lemmas (extensive, monotone, idempotent) are proved in proofs/abstract_class_system.lean via AxisClosure. Exchange and equicardinality lemmas are proved in proofs/axis_framework.lean via L4 and L1. The composition point is formalized by closureInducedAxisMatroid and L1, with closure-induced nodup/subset/exchange hypotheses made explicit.

Definition IV.6 (Bases family)**.** Let $E = \Phi$ $(= \mathcal{Q})$ be the ground set of primitive queries (attribute predicates). Let $\mathcal{B} \subseteq 2^E$ be the family of minimal distinguishing sets.

Lemma IV.7 (Basis exchange)**.** *For any $B_1, B_2 \in \mathcal{B}$ and any $q \in B_1 \setminus B_2$, there exists $q' \in B_2 \setminus B_1$ such that $(B_1 \setminus \{q\}) \cup \{q'\} \in \mathcal{B}$.*

(L: L4, L5)

Proof. Define the closure operator $\mathrm{cl}(X) = \{q : X\text{-equivalence implies } q\text{-equivalence}\}$. We verify the matroid axioms:

1) **Closure axioms**: cl is extensive, monotone, and idempotent. These follow directly from the definition of logical implication.
2) **Exchange property**: If $q \in \mathrm{cl}(X \cup \{q'\}) \setminus \mathrm{cl}(X)$, then $q' \in \mathrm{cl}(X \cup \{q\})$.

For exchange, take $q \in \mathrm{cl}(X \cup \{q'\}) \setminus \mathrm{cl}(X)$. Since $q \notin \mathrm{cl}(X)$, there exist v, w that are X-equivalent but disagree on q. Because $q \in \mathrm{cl}(X \cup \{q'\})$, any pair that is $(X \cup \{q'\})$-equivalent must agree on q; therefore this witness pair cannot be $(X \cup \{q'\})$-equivalent, so it must disagree on q'. Now fix any pair v', w' that are $(X \cup \{q\})$-equivalent. They are in particular X-equivalent and agree on q. If they disagreed on q', then by the previous implication we could derive disagreement on q, contradiction. Hence v', w' agree on q', proving $q' \in \mathrm{cl}(X \cup \{q\})$.

Minimal distinguishing sets are exactly the bases of the matroid defined by this closure operator. The closure component is machine-checked in `AxisClosure`; the exchange/equicardinality component and the closure-to-matroid bridge used for distinguishing-dimension well-definedness are machine-checked in `axis_framework.lean`. ∎

Theorem IV.8 (Matroid bases)**.** *$\mathcal{B}$ is the set of bases of a matroid on ground set E.*

(L: L1, L5)

Proof. By the basis-exchange lemma and the standard characterization of matroid bases [12]. ∎

Definition IV.9 (Distinguishing dimension)**.** The *distinguishing dimension* of a classification system is the common cardinality of all minimal distinguishing sets.

Remark IV.10 (Ambient attribute count vs. distinguishing dimension). Let $n := |\mathcal{I}|$ be the ambient number of available attributes. Clearly $d \leq n$, and there exist worst-case families with $d = n$.

Corollary IV.11 (Well-defined distinguishing dimension)**.** *All minimal distinguishing sets have equal cardinality. Thus the distinguishing dimension (Definition IV.9) is well-defined.*

(L: L1)

D. Implications for Witness Cost

Corollary IV.12 (Lower bound on attribute-only witness cost)**.** *For any attribute-only observer, $W(\textit{class-identity}) \geq d$ where d is the distinguishing dimension.*

(L: ACS1)

Proof. If a procedure queried fewer than d attributes on every execution path, each such queried set would be non-distinguishing by definition of d. For that path, there would exist two different classes with identical answers on all queried attributes, yielding identical transcripts and forcing the same output on both values. This contradicts zero-error class identification. Hence some path requires at least d queries. ■

The key insight: the distinguishing dimension is invariant across all minimal query strategies. The difference between nominal-tag and attribute-only observers lies in *witness cost*: a nominal tag achieves $W = O(1)$ by storing the identity directly, bypassing query enumeration.

V. Witness Cost Analysis

A. Witness Cost for Class Identity

Recall from Section 2 that the witness cost $W(P)$ is the minimum number of primitive queries required to compute property P. For class identity, we ask: what is the minimum number of queries to determine if two values have the same class?

Theorem V.1 (Nominal-Tag Observers Achieve Minimum Witness Cost)**.** *Nominal-tag observers achieve the minimum witness cost for class identity:*

$$W_{eq} = O(1)$$

Specifically, the witness is a single tag read: compare tag(v_1) = tag(v_2).

Attribute-only observers require $W_{eq} = \Omega(d)$ where d is the distinguishing dimension (and $d \leq n$, with worst-case $d = n$).

(L: ACS7, ACS1)

Proof. See Lean formalization: `proofs/nominal_resolution.lean`. The proof shows:

1) Nominal-tag access is a single primitive query
2) Attribute-only observers must query at least d attributes in the worst case (a generic strategy queries all n)
3) No shorter witness exists for attribute-only observers (by the information barrier)

■

B. Witness Cost Comparison

Observer Class	Witness Procedure	Witness Cost W
Nominal-tag	Single tag read	$O(1)$
Attribute-only	Query a distinguishing set	$\Omega(d)$

TABLE II
WITNESS COST FOR CLASS IDENTITY BY OBSERVER CLASS.

The Lean 4 formalization in the supplementary artifact provides a machine-checked proof that nominal-tag access minimizes witness cost for class identity.

VI. (L, W, D) OPTIMALITY

A. Three-Dimensional Tradeoff: Tag Length, Witness Cost, Distortion

Recall from Section 2 that observer strategies are characterized by three dimensions:

- **Tag length** L: bits required to encode a class identifier ($L \geq \log_2 k$ for k classes under full class tagging)
- **Witness cost** W: minimum number of primitive queries for class identification
- **Distortion** D: probability of misclassification, $D = \Pr[\hat{C} \neq C]$.

We compare two observer classes:

Definition VI.1 (Attribute-only observer)**.** An observer that queries only attribute membership ($q_I \in \Phi_{\mathcal{I}}$), with no access to explicit class tags.

Definition VI.2 (Nominal-tag observer)**.** An observer that may read a single class identifier (nominal tag) per value, in addition to attribute queries.

Theorem VI.3 (Pareto Optimality of Nominal-Tag Observers)**.** *Let A_π be the collision multiplicity (Definition II.28). Then:*

1) *Any $D = 0$ scheme must satisfy $L \geq \log_2 A_\pi$ (Theorem II.36).*
2) *In maximal-barrier domains ($A_\pi = k$), nominal-tag observers achieve the unique Pareto-optimal $D = 0$ point:*
   - ***Tag length**: $L = \lceil \log_2 k \rceil$ bits for k classes*
   - ***Witness cost**: $W = O(1)$ queries (one tag read)*
   - ***Distortion**: $D = 0$ (zero misclassification probability)*
3) *In general (non-maximal) domains, nominal tagging remains Pareto-optimal at $D = 0$ but need not be unique: partial tags can coexist on the frontier.*

In information-barrier domains, attribute-only observers (the $L = 0$ face) satisfy:

- ***Tag length**: $L = 0$ bits (no explicit tag)*
- ***Witness cost**: $W = \Omega(d)$ queries (must query at least one minimal distinguishing set of size d, see Definition IV.9)*
- ***Distortion**: $D > 0$ (probability of misclassification is strictly positive due to collisions)*

(L: LWDC1, LWDC3, LWDC2)

Proof. Converse item (1) is Theorem II.36. For item (2), maximal barrier means all classes are observationally colliding, so any $D = 0$ scheme must carry full class identity in tag bits (Corollary II.37), while nominal tags realize this lower bound with one tag read. Pareto uniqueness follows because any competing $D = 0$ point cannot reduce L below $\log_2 k$ nor reduce W below constant-time tag access under the admissibility rules of Section II.

The converse proof path is machine-checked in Lean: `proofs/lwd_converse.lean` formalizes (i) constant transcript on a collision block implies tag injectivity under zero-error decoding, and (ii) counting then yields $2^L \geq A_\pi$. The maximal-barrier corollary is formalized

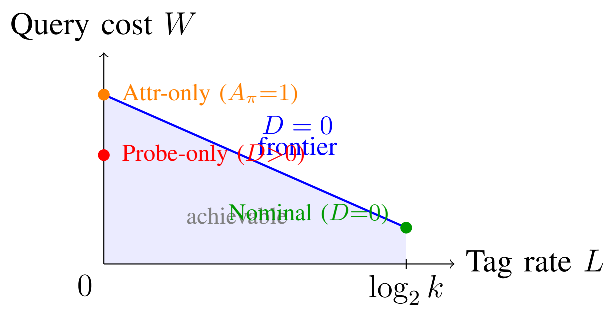

Fig. 1. Schematic (L, W, D) tradeoff across barrier regimes. Nominal tagging is the unique zero-error Pareto point in maximal-barrier domains; attribute-only strategies lie on the zero-tag face and leave the zero-error frontier when collisions are present.

in the same module. Runtime cost instantiations (e.g., unbounded gap examples) remain in `proofs/python_instantiation.lean`. ■

Remark VI.4 (General $D = 0$ Frontier). When $1 < A_\pi < k$, the $D = 0$ frontier can include mixed designs: a partial tag identifies collision blocks and queries resolve within blocks. This does not contradict nominal optimality; it only removes global uniqueness outside maximal-barrier domains.

1) Maximal barrier ($A_\pi = k$): unique $D = 0$ nominal point.
2) Intermediate barrier ($1 < A_\pi < k$): multiple $D = 0$ Pareto points may exist.
3) No barrier ($A_\pi = 1$): $L = 0$ zero-error identification is feasible.

B. Pareto Frontier

The three-dimensional frontier shows:

- In maximal-barrier domains, the unique $D = 0$ Pareto point is nominal tagging at $L = \lceil \log_2 k \rceil$.
- In general domains, attribute-only observers trade tag length for distortion on the $L = 0$ face when collisions are present.

Figure 1 visualizes the (L, W, D) tradeoff space. The key observation is the ambiguity converse: the minimum zero-error tag rate is $\log_2 A_\pi$, with the maximal-barrier special case $\log_2 k$.

The Lean 4 formalization in the supplementary artifact machine-checks the full ambiguity-based converse chain and maximal-barrier lower bound that anchor this tradeoff analysis.

Remark VI.5 (Software-runtime instantiation (example)). In one software-runtime instantiation: explicit runtime identifiers correspond to nominal-tag observers (e.g., CPython's `isinstance`, Java's `.getClass()`); attribute-probe schemes correspond to attribute-only observers (e.g., repeated `hasattr`-style checks); and interface-conformance checks are an intermediate case with $D = 0$ but $W = O(s)$.

Remark VI.6 (Conformance-check cost parameter). When conformance checks traverse s members/fields (rather than ranging over the full attribute universe), the natural bound is $W = O(s)$ with $s \leq n$.

VII. Cross-Domain Instantiations

This section gives brief illustrations showing the same constraint pattern in established systems.

A. Compact Cross-Domain Illustrations

Biology (phenotype vs genotype). Distinct species can collide under phenotype, so phenotype-only observation cannot guarantee zero-error identity. DNA barcoding supplies additional naming information that resolves collision blocks in constant-time lookup style [13].

Databases (columns vs keys). Rows may collide on non-key columns, so attribute-only identification is insufficient for guaranteed identity. Primary/surrogate keys realize the nominal-tag role and provide constant-cost identity checks [14].

Software runtimes (attribute probes vs runtime identifiers). Attribute-probe checks recover identity from observed capabilities and pay query cost that scales with inspected structure. Runtime class/type identifiers behave as explicit tags and collapse identity checks to near-constant cost in the model [15]–[19].

B. Takeaway

These illustrations are not historical causality claims. They show a common resource law: when observable profiles collide, zero-error identification requires additional naming information; otherwise the system pays higher query cost and/or nonzero error.

VIII. Conclusion

This paper presents a resource-based analysis of classification under observational constraints. We establish three core theorem families and accompanying extensions:

1) **Information Barrier**: Observers limited to attribute-membership queries cannot compute properties that vary within indistinguishability classes. This is the structural obstruction from which the later lower bounds and tradeoffs follow.
2) **Witness Optimality**: Nominal-tag observers achieve $W(\text{identity}) = O(1)$, the minimum witness cost. The gap from attribute-only observation ($\Omega(d)$, with a worst-case family where $d = n$) is unbounded.
3) **Matroid Structure**: Minimal distinguishing query sets form the bases of a matroid. The distinguishing dimension of a classification problem is well-defined and computable.

A. A Recurring Design Constraint

Across domains, the same structure recurs:

- **Biology**: Phenotypic observation cannot distinguish cryptic species. DNA barcoding (nominal tag) resolves them in $O(1)$.
- **Databases**: Column-value queries cannot distinguish rows with identical attributes. Primary keys (nominal tag) provide $O(1)$ identity.
- **Software runtimes**: Attribute observation cannot distinguish entities that collide on observed capabilities. Runtime identifiers provide $O(1)$ identity.

The information barrier is not a quirk of any particular domain; it is a mathematical necessity arising from the quotient structure induced by limited observations. What varies by domain is not the existence of the constraint, but which resource the system chooses to spend to escape it.

B. Implications

- **The necessity of nominal information is a theorem, not a preference.** Any zero-error scheme must satisfy the ambiguity converse $L \geq \log_2 A_\pi$ (Theorem II.36), where A_π is the largest collision block induced by observable profiles. In maximal-barrier domains ($A_\pi = k$), this becomes $L \geq \log_2 k$ and nominal tagging gives the unique $D = 0$ Pareto point with $W = O(1)$ (Theorem VI.3).
- **The barrier is informational, not computational**: even with unbounded resources, attribute-only observers cannot overcome it.
- **Finite-block confusability laws are exact**: the mechanized graph-confusability results give one-shot and block thresholds (A_π and A_π^k) and exact linear log-bit scaling across block length.
- **Open-world guarantees require explicit tags**: barrier-freedom is not extension-stable under system growth, and deciding barrier existence/freedom for arbitrary generators is not computable (Rice-style). So a persistent $D = 0$ guarantee cannot rely on "currently collision-free" attribute structure alone.
- **Fixed-axis systems are necessarily incomplete outside their axis**: by Corollary II.6, any fixed-axis classifier is complete only for axis-measurable properties and cannot represent properties that vary within an axis fiber unless a new axis or explicit tag is introduced.
- **Classification system design is constrained**: the choice of observation family determines which properties are computable.

(L: ROB1, ROB2, UND1, UND2)

C. Future Work

Natural extensions include other classification domains, witness complexity for properties beyond identity, and hybrid observers that combine tags and attributes under non-uniform query distributions.

D. Conclusion

Classification under observational constraints admits a clean quantitative analysis. The zero-error converse is governed by collision multiplicity: any $D = 0$ scheme necessarily has $L \geq \log_2 A_\pi$ (Theorem II.36). In maximal-barrier domains ($A_\pi = k$), nominal-tag observation achieves the unique Pareto-optimal $D = 0$ point in the (L, W, D) tradeoff (Theorem VI.3). The mechanized graph-confusability extension further supplies exact finite-block thresholds and rate scaling. Within the stated observation model, the paper turns a familiar heuristic into explicit lower bounds, explicit dominance claims, and structural invariants, and all proofs are machine-verified in Lean 4.

Acknowledgment: AI-use Disclosure

Generative AI tools (including Codex, Claude Code, Augment, Kilo, and OpenCode) were used throughout this manuscript, across all sections (Abstract, Introduction, theoretical development, proof sketches, applications, conclusion, and appendix) and across all stages from initial drafting to final revision. The tools were used for boilerplate generation, prose and notation refinement, LaTeX/structure cleanup, translation of informal proof ideas into candidate formal artifacts (Lean/LaTeX), and repeated adversarial reviewer-style critique passes to identify blind spots and clarity gaps.

The author retained full intellectual and editorial control, including problem selection, theorem statements, assumptions, novelty framing, acceptance criteria, and final inclusion/exclusion decisions. No technical claim was accepted solely from AI output. Formal claims reported as machine-verified were admitted only after Lean verification (no `sorry` in cited modules) and direct author review; Lean was used as an integrity gate for responsible AI-assisted research. The author is solely responsible for all statements, citations, and conclusions.

References

[1] R. Ahlswede and G. Dueck, "Identification via channels," *IEEE Transactions on Information Theory*, vol. 35, no. 1, pp. 15–29, 1989.

[2] ——, “Identification in the presence of feedback—a discovery of new capacity formulas,” *IEEE Transactions on Information Theory*, vol. 35, no. 1, pp. 30–36, 1989.

[3] C. E. Shannon, “Coding theorems for a discrete source with a fidelity criterion,” *IRE National Convention Record*, vol. 7, pp. 142–163, 1959.

[4] T. Berger, *Rate Distortion Theory: A Mathematical Basis for Data Compression*. Prentice-Hall, 1971.

[5] Y. Blau and T. Michaeli, “Rethinking lossy compression: The rate-distortion-perception tradeoff,” in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 675–685. [Online]. Available: https://proceedings.mlr.press/v97/blau19a.html

[6] J. Körner, “Coding of an information source having ambiguous alphabet and the entropy of graphs,” *6th Prague Conference on Information Theory*, pp. 411–425, 1973.

[7] H. S. Witsenhausen, “The zero-error side information problem and chromatic numbers,” *IEEE Transactions on Information Theory*, vol. 22, no. 5, pp. 592–593, 1976.

[8] H. Buhrman and R. de Wolf, “Complexity measures and decision tree complexity: A survey,” *Theoretical Computer Science*, vol. 288, no. 1, pp. 21–43, 2002.

[9] A. Orlitsky, “Worst-case interactive communication i: Two messages are almost optimal,” *IEEE Transactions on Information Theory*, vol. 36, no. 5, pp. 1111–1126, 1990.

[10] L. Cardelli and P. Wegner, “On understanding types, data abstraction, and polymorphism,” *ACM Computing Surveys*, vol. 17, no. 4, pp. 471–523, 1985.

[11] W. R. Cook, W. Hill, and P. S. Canning, “Inheritance is not subtyping,” in *Proceedings of the 17th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, 1990, pp. 125–135.

[12] D. J. A. Welsh, *Matroid Theory*. Academic Press, 1976.

[13] P. D. N. Hebert, A. Cywinska, S. L. Ball, and J. R. deWaard, “Biological identifications through dna barcodes,” *Proceedings of the Royal Society of London. Series B: Biological Sciences*, vol. 270, no. 1512, pp. 313–321, 2003.

[14] E. F. Codd, *The Relational Model for Database Management: Version 2*. Addison-Wesley, 1990.

[15] P. S. Foundation, “Cpython object implementation: Object structure,” https://github.com/python/cpython/blob/main/Include/object.h, 2024.

[16] Oracle, “The java virtual machine specification: Chapter 2.5 runtime data areas,” https://docs.oracle.com/javase/specs/jvms/se21/html/jvms-2.html, 2024.

[17] ——, “Java object: getclass(),” https://docs.oracle.com/javase/8/docs/api/java/lang/Object.html#getClass--, 2024.

[18] Microsoft, “Typescript handbook: Type compatibility,” https://www.typescriptlang.org/docs/handbook/type-compatibility.html, 2024.

[19] Rust Project Developers, “Rust standard library: `std::any::TypeId`,” https://doc.rust-lang.org/std/any/struct.TypeId.html, 2024.

Lean Handle Index.

ID	Handle	ID	Handle
ACS1	adversary_forces_n_minus_1_queries	ACS2	complexity_gap_unbounded
ACS3	duck_localization_linear	ACS4	encoding_location_gap

ID	Handle	ID	Handle
ACS5	model_completeness	ACS6	nominal_centralized
ACS7	nominal_localization_constant_semantic	ACS8	shape_cannot_distinguish
ACS9	shape_provenance_impossible	COH1	preference_incoherent
COMPL1	axes_complete	DOM1	strict_dominance
FORC2	typing_choice_unavoidable	GPH13	Ssot.GraphEntropy.tagFeasible_iff_maxFiberCard_le
GPH16	Ssot.GraphEntropy.block_tagFeasible_iff_pow_maxFiberCard_le	GPH17	Ssot.GraphEntropy.maxFiberCard_blockObserve_eq_pow
GPH18	Ssot.GraphEntropy.blockTagRateBits_eq_mul_oneShot	GPH19	Ssot.GraphEntropy.blockTagRateBitsPerCoordinate_eq_oneShot
GPH20	Ssot.GraphEntropy.minBlockFeasibleAlphabet_eq_pow	GPH24	Ssot.GraphEntropy.feasibleAtAlphabetBase_iff
L1	matroid_basis_equicardinality	L2	fixed_axis_incompleteness
L4	l4_exchange_wrapper	L5	l5_exchange_wrapper
LWDC1	LWDConverse.collision_block_requires_bits	LWDC2	LWDConverse.impossible_when_bits_too_small
LWDC3	LWDConverse.maximal_barrier_requires_bits	ROB1	AbstractClassSystem.OpenWorld.extend_to_force_barrier
ROB2	AbstractClassSystem.OpenWorld.barrierFreedom_not_extension_stable	UND1	AbstractClassSystem.OpenWorld.hasBarrier_not_computable
UND2	AbstractClassSystem.OpenWorld.barrierFree_not_computable	UNIQ1	unique_tree_encoding